

MATERNAL RISK PREDICTION

Data Mining Based Classification Analysis

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Abstract

Maternal health risk assessment is a critical healthcare problem that directly affects both maternal and fetal well-being. Early identification of high-risk pregnancies enables healthcare professionals to provide timely medical intervention, continuous monitoring, and preventive treatment strategies. However, manually analyzing large-scale medical data can be difficult, time-consuming, and susceptible to human error. Therefore, data mining techniques have become increasingly important in healthcare-oriented decision support systems.

This project focuses on maternal risk prediction using supervised data mining classification algorithms applied to a structured maternal health dataset. The dataset contains physiological and medical indicators such as blood pressure, blood sugar level, body mass index (BMI), heart rate, age, diabetes-related conditions, and previous pregnancy complications. The objective of the study is to classify pregnant individuals into low-risk and high-risk categories based on these medical measurements.

A complete data mining workflow was implemented, including exploratory data analysis (EDA), missing value handling, feature scaling, target encoding, model training, performance evaluation, and cross-validation analysis. Multiple classification algorithms were evaluated, including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest.

Experimental results demonstrated that all classification models achieved strong predictive performance on the maternal health dataset. Among the evaluated approaches, the Random Forest algorithm achieved the highest overall performance with an accuracy of 98.34% and a ROC-AUC score of 0.9863. The findings also revealed that medical indicators such as blood sugar level, preexisting diabetes, and BMI play significant roles in maternal risk prediction.

The results of this project demonstrate that data mining techniques can effectively support maternal healthcare analytics and may contribute to the development of reliable clinical decision-support systems for early maternal risk detection.

1. Introduction

Maternal health is one of the most critical areas of modern healthcare systems because complications during pregnancy can directly affect both maternal and fetal survival. According to global healthcare studies, early identification of pregnancy-related risks significantly reduces maternal mortality rates and improves overall pregnancy outcomes. Therefore, accurate maternal risk assessment has become an essential component of preventive healthcare and clinical decision-support systems.

Traditional risk assessment methods often rely on manual interpretation of medical measurements and clinical expertise. However, modern healthcare environments generate large volumes of medical data, making manual analysis increasingly difficult, time-consuming, and prone to human error. As a result, data mining techniques have become highly valuable for extracting meaningful patterns from healthcare datasets and supporting medical decision-making processes.

Data mining classification algorithms can analyze complex relationships between medical indicators and disease outcomes, allowing predictive models to identify high-risk patients with high accuracy. In maternal healthcare applications, variables such as blood pressure, blood sugar level, body mass index (BMI), heart rate, diabetes-related conditions, and previous pregnancy complications may strongly influence pregnancy risk levels. Discovering hidden relationships among these variables can help healthcare professionals perform earlier interventions and improve patient monitoring processes.

The primary objective of this project is to develop and evaluate multiple supervised data mining classification models for maternal risk prediction using a structured maternal health dataset. The dataset contains physiological and medical measurements collected from pregnant individuals together with maternal risk labels representing low-risk and high-risk pregnancy conditions.

To achieve this objective, a complete data mining workflow was implemented throughout the project. The study includes exploratory data analysis (EDA), missing value handling, target encoding, feature scaling, train-test splitting, model training, cross-validation analysis, and performance evaluation. Several classification algorithms were implemented and compared, including:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

The models were evaluated using multiple performance metrics such as accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and cross-validation scores. In addition to prediction performance, the study also aims to analyze the contribution of different medical indicators to maternal risk classification.

The experimental findings demonstrate that data mining approaches can successfully classify maternal health risks with high predictive accuracy. Among the evaluated models, ensemble-based approaches achieved the strongest overall performance, highlighting the effectiveness of advanced classification techniques in healthcare analytics.

The remainder of this report is organized as follows:

- Section 4 presents the background information and related studies associated with maternal risk prediction and healthcare-oriented classification systems.
- Section 5 explains the algorithms and methodology used throughout the project.
- Section 6 describes the experimental setup, dataset characteristics, preprocessing operations, and implementation environment.
- Section 7 presents the experimental evaluation results and comparative performance analysis of all classification models.
- Finally, Section 8 summarizes the conclusions of the study and discusses potential future improvements.

2. Background and Related Work

Maternal health risk prediction has become an increasingly important research area in healthcare analytics and data mining due to the growing availability of digital medical records and clinical datasets. Early detection of pregnancy-related complications is essential for reducing maternal mortality, improving fetal health outcomes, and supporting preventive healthcare systems. As a result, researchers have widely explored the use of data mining and classification techniques for maternal risk assessment and clinical decision-support applications.

Traditional maternal risk assessment methods are generally based on manual medical evaluation, laboratory testing, and physician experience. Although these approaches are clinically valuable, they may become insufficient when analyzing large-scale healthcare data containing multiple physiological and medical indicators. Data mining techniques provide automated and data-driven solutions capable of identifying hidden relationships and predictive patterns within medical datasets.

Several previous studies have investigated the use of classification algorithms in healthcare-oriented prediction problems. Commonly used approaches include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

- Artificial Neural Networks

These algorithms have been widely applied in disease diagnosis, patient risk stratification, diabetes prediction, cardiovascular disease detection, and maternal healthcare analytics.

Among traditional classification algorithms, Logistic Regression is frequently preferred because of its simplicity, interpretability, and strong performance in binary classification tasks. In healthcare applications, the probabilistic outputs generated by Logistic Regression can support clinical interpretation and decision-making processes.

K-Nearest Neighbors (KNN) has also been widely used in medical classification studies because it classifies patients based on similarity between medical records. Since patients with similar physiological characteristics may exhibit similar disease patterns, KNN can effectively capture local relationships within structured healthcare datasets.

Support Vector Machine (SVM) is another commonly used classification approach in medical prediction systems. SVM is particularly effective in high-dimensional classification problems because it maximizes the separation margin between classes and provides strong generalization capability. Previous healthcare studies have demonstrated that SVM can achieve highly competitive performance in disease classification tasks.

Tree-based algorithms such as Decision Tree and Random Forest have gained significant popularity in healthcare analytics due to their interpretability and ability to model nonlinear relationships. Decision Trees provide transparent decision structures that allow medical professionals to understand classification rules more easily. Random Forest further improves prediction performance by combining multiple decision trees using ensemble learning techniques, reducing overfitting risk and improving model robustness.

Recent healthcare-oriented data mining studies have shown that ensemble-based approaches generally outperform simpler classification algorithms in structured tabular datasets. In particular, Random Forest models have demonstrated strong success in medical risk prediction problems due to their stability, feature importance analysis capability, and high classification accuracy.

Previous maternal healthcare studies have identified several important predictors associated with pregnancy-related risks, including:

- blood sugar level (BS),
- systolic and diastolic blood pressure,
- body mass index (BMI),
- diabetes-related conditions,
- maternal age,
- and previous pregnancy complications.

These medical indicators are strongly associated with conditions such as gestational diabetes, hypertension, obesity-related complications, and cardiovascular abnormalities during pregnancy.

The dataset used in this project contains many of these clinically important medical variables, making it highly suitable for healthcare-oriented classification analysis. Unlike studies that focus on a single classification algorithm, this project provides a comparative evaluation of multiple supervised data mining models using the same maternal health dataset and evaluation framework.

The findings reported in previous literature strongly support the effectiveness of data mining techniques in healthcare analytics. Motivated by these studies, this project aims to evaluate and compare multiple classification approaches in order to identify the most reliable and accurate maternal risk prediction model.

3. Algorithms and Methodology

This project applies supervised data mining classification techniques to predict maternal health risk levels using physiological and medical indicators. The primary objective of the methodology is to identify patterns within maternal healthcare data and develop predictive models capable of accurately classifying pregnant individuals into low-risk and high-risk categories.

The overall methodology of the project consists of several sequential stages, including:

1. Data preparation and preprocessing
2. Exploratory Data Analysis (EDA)
3. Feature preparation and transformation
4. Classification model implementation
5. Performance evaluation and comparative analysis
6. Knowledge discovery and interpretation

A complete data mining workflow was implemented to ensure reliable, reproducible, and statistically consistent experimental results.

3.1 Data Mining Approach

This study focuses on supervised classification, which is one of the most widely used data mining techniques in healthcare analytics. In supervised classification problems, predictive models learn the relationship between input features and predefined target labels using previously labeled observations.

The classification process can generally be represented as:

$$\hat{y} = f(X)$$

where:

- X represents the input medical features,
- f represents the learned machine learning model,
- and $\hat{y}$ represents the predicted maternal risk level.

The target variable of this project is binary:

- Low Risk (0)
- High Risk (1)

Several supervised classification algorithms were implemented and compared throughout the study.

3.2 Selected Classification Algorithms

Five different classification algorithms were used in this project:

Logistic Regression

Logistic Regression is a probability-based linear classification algorithm commonly used in binary prediction problems. The algorithm estimates the probability that an observation belongs to a specific class using the sigmoid function.

Logistic Regression was selected because:

- it performs effectively on binary classification tasks,
- provides interpretable probability outputs,
- and offers computational efficiency.

Its interpretability makes it particularly suitable for healthcare-oriented prediction systems.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a distance-based classification algorithm that predicts class labels according to the nearest neighboring observations within the feature space.

KNN was selected because:

- it captures similarity-based relationships between patients,
- performs effectively on structured datasets,
- and can model local patterns within healthcare data.

Since KNN depends heavily on distance calculations, feature scaling was applied before training.

The Euclidean distance formula used in KNN is:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Support Vector Machine (SVM)

Support Vector Machine (SVM) is a margin-based classification algorithm designed to maximize the separation boundary between classes.

SVM was selected because:

- it performs strongly on high-dimensional datasets,
- provides strong generalization capability,
- and effectively separates complex class boundaries.

The decision boundary of SVM can be represented as:

$$w^T x + b = 0$$

Feature standardization was particularly important for SVM because the algorithm is highly sensitive to feature magnitudes.

Decision Tree

Decision Tree is a hierarchical rule-based classification algorithm that recursively partitions the dataset using feature-based decision rules.

Decision Trees were selected because:

- they are highly interpretable,
- capable of capturing nonlinear relationships,
- and require minimal statistical assumptions.

Decision Tree models are especially useful in healthcare applications because their decision paths can be interpreted clinically.

The impurity measure used in tree splitting is based on Gini Impurity:

$$Gini = 1 - \sum_{i=1}^c p_i^2$$

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification performance and reduce overfitting.

Random Forest was selected because:

- it provides high predictive accuracy,
- improves model robustness,
- handles nonlinear relationships effectively,

- and supports feature importance analysis.

The final prediction mechanism is based on majority voting across multiple trees:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_n(x))$$

Among all evaluated algorithms, Random Forest demonstrated the strongest overall classification performance.

3.3 Data Preparation Workflow

Before model training, several preprocessing operations were applied to transform the raw maternal health dataset into a machine-learning-compatible format.

The preprocessing workflow included:

- missing value handling,
- target encoding,
- feature-target separation,
- train-test splitting,
- and feature scaling.

Missing numerical values were handled using median imputation in order to reduce the influence of outliers. Categorical target values were handled using mode imputation.

The maternal risk labels were encoded as:

- Low Risk $\rightarrow$ 0
- High Risk $\rightarrow$ 1

After preprocessing, the dataset was divided into training and testing subsets using an 80%-20% split ratio.

3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted before model training in order to better understand the structure and characteristics of the dataset.

The EDA stage included:

- missing value analysis,
- feature distribution analysis,
- outlier detection,
- correlation analysis,
- and risk-group-based comparisons.

Several visualization techniques were used throughout the analysis process, including:

- histograms,
- boxplots,
- QQ plots,
- and correlation heatmaps.

The EDA findings revealed that variables such as blood sugar level, blood pressure, BMI, and diabetes-related indicators have strong relationships with maternal risk levels.

The analysis also demonstrated the presence of medically meaningful outliers, which were preserved because they may represent clinically significant high-risk observations rather than data collection errors.

3.5 Feature Selection and Knowledge Discovery

The project focuses on identifying medically significant predictors associated with maternal risk classification.

Feature relationships were analyzed using:

- correlation analysis,
- comparative visualization techniques,
- and model-based importance evaluation.

The knowledge discovery process revealed that several medical indicators strongly contribute to maternal risk prediction, particularly:

- blood sugar level (BS),
- preexisting diabetes,
- BMI,
- systolic blood pressure,
- and previous pregnancy complications.

These findings are consistent with previous healthcare studies related to maternal risk assessment and pregnancy complications.

3.6 Performance Evaluation Metrics

Multiple evaluation metrics were used to comprehensively analyze classification performance because a single metric alone may not fully represent model effectiveness.

The evaluated metrics include:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

- Cross Validation Scores

Accuracy measures overall classification success and is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the reliability of positive predictions:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures how effectively high-risk maternal cases are identified:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall into a single balanced metric:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC-AUC analysis was also used to evaluate class discrimination capability across all classification models.

Additionally, 5-Fold Cross Validation was applied to measure model generalization performance and reduce evaluation bias.

4. Experimental Setup

This section describes the experimental environment, dataset structure, preprocessing operations, and implementation strategy used throughout the maternal risk prediction project. The experimental setup was designed to ensure reproducible, reliable, and statistically consistent classification results.

4.1 Dataset Description

The dataset used in this project contains maternal health-related medical measurements collected for pregnancy risk assessment and classification analysis. The primary objective of the dataset is to predict maternal risk levels using physiological and medical indicators associated with pregnancy complications.

The dataset consists of:

- 1205 observations (samples)
- 12 attributes (columns)

Among these attributes:

- 11 columns represent input medical features,
- and 1 column represents the target variable (Risk Level).

The target variable contains two binary classes:

- Low Risk
- High Risk

For classification modeling, the target labels were encoded numerically as:

<i>Original Label</i>	<i>Encoded Value</i>
<i>Low</i>	0
<i>High</i>	1

The dataset contains both continuous numerical variables and binary medical-condition indicators. The features included in the analysis are summarized below:

<i>Feature Name</i>	<i>Description</i>
<i>Age</i>	Age of the pregnant individual
<i>Systolic BP</i>	Systolic blood pressure
<i>Diastolic</i>	Diastolic blood pressure
<i>BS</i>	Blood sugar level
<i>Body Temp</i>	Body temperature
<i>BMI</i>	Body Mass Index
<i>Previous Complications</i>	Previous pregnancy complications
<i>Preexisting Diabetes</i>	Diabetes before pregnancy
<i>Gestational Diabetes</i>	Diabetes during pregnancy
<i>Mental Health</i>	Mental health condition indicator
<i>Heart Rate</i>	Heart rate measurement
<i>Risk Level</i>	Maternal risk category

The dataset includes several medically significant variables strongly associated with maternal healthcare analytics, including cardiovascular measurements, metabolic indicators, diabetes-related conditions, and physiological observations.

Although the dataset is not perfectly balanced, the class distribution remains sufficiently suitable for supervised classification modeling. The Low Risk category contains a larger number of observations than the High Risk category; however, the imbalance is not severe enough to require resampling techniques.

Several medically meaningful outliers were identified during analysis, particularly in:

- blood sugar level (BS),
- BMI,
- systolic blood pressure,
- and diastolic blood pressure.

These outliers were preserved because they may represent clinically important high-risk pregnancy conditions rather than data collection errors.

4.2 Preprocessing Steps

Before model training, several preprocessing operations were applied to prepare the dataset for classification analysis. The preprocessing pipeline was implemented programmatically using Python and Scikit-learn libraries.

The preprocessing workflow included:

1. Missing value handling
2. Target encoding
3. Feature-target separation
4. Train-test splitting
5. Feature scaling

Missing Value Handling

The dataset contained missing observations in several medical variables, including BMI, blood pressure measurements, and the target column.

Instead of removing incomplete observations, imputation techniques were applied to preserve dataset size and maintain statistical consistency.

Numerical missing values were handled using median imputation because medical datasets often contain skewed distributions and outliers. The median is more robust than the mean in such cases.

Target variable missing values were handled using mode imputation to preserve class consistency within the dataset.

Target Encoding

The categorical maternal risk labels were transformed into numerical values to make the dataset compatible with supervised classification algorithms.

The encoding strategy was implemented as:

- Low Risk → 0
- High Risk → 1

This transformation converted the problem into a binary classification task.

Feature and Target Separation

After preprocessing and encoding operations, the dataset was separated into:

- feature matrix (X)
- and target vector (y).

The feature matrix contains all medical indicators used for prediction, while the target vector represents maternal risk labels.

Feature Scaling

Since the dataset contains variables operating on different numerical scales, feature standardization was applied before model training.

Algorithms such as:

- KNN,
- Logistic Regression,
- and SVM

are highly sensitive to feature magnitudes; therefore, StandardScaler was used to normalize the feature distributions.

The standardization formula is:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = original feature value,
- μ = mean of the feature,
- σ = standard deviation,
- z = standardized value.

Feature scaling improved:

- numerical stability,
- optimization performance,
- and distance-based learning quality.

4.3 Experimental Design

The experimental design was constructed to evaluate the predictive capability and generalization performance of multiple classification algorithms under consistent conditions.

Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

using the Scikit-learn `train_test_split()` function with:

$$random_state = 42$$

This fixed random seed ensured reproducibility and experimental consistency across multiple executions.

After splitting:

- the training dataset contained 964 observations,
- while the testing dataset contained 241 observations.

Cross Validation Strategy

To evaluate model generalization capability and reduce evaluation bias, 5-Fold Cross Validation was applied during the evaluation stage.

The dataset was repeatedly partitioned into five subsets:

1. Four subsets were used for training.
2. One subset was used for validation.
3. The process repeated until every subset had been used once as validation data.

The mean validation score was calculated using:

$$CV_{mean} = \frac{1}{k} \sum_{i=1}^k Score_i$$

Cross Validation improved:

- statistical reliability,
- robustness analysis,
- and overfitting detection capability.

Hyperparameter Configuration

Several classification algorithms were implemented using predefined parameter settings.

The main configurations included:

Algorithm	Configuration
<i>Logistic Regression</i>	random_state=42
<i>KNN</i>	n_neighbors=5
<i>SVM</i>	kernel='rbf'
<i>Decision Tree</i>	random_state=42
<i>Random Forest</i>	n_estimators=100

These configurations were implemented directly within the training pipeline.

4.4 Software and Hardware Environment

The entire project was implemented using Python-based data mining and machine learning libraries.

The experimental environment included:

<i>Component</i>	<i>Description</i>
<i>Programming Language</i>	Python
<i>Development Environment</i>	Jupyter Notebook
<i>Data Processing</i>	Pandas
<i>Numerical Computation</i>	NumPy
<i>Visualization</i>	Matplotlib, Seaborn
<i>Machine Learning Library</i>	Scikit-learn

The project pipeline was modularly organized into separate preprocessing, training, and evaluation scripts, including:

- `preprocessing.py`
- `train_models.py`
- `evaluation.py`

The experiments were executed in a standard local computing environment without GPU acceleration because the dataset size and model complexity did not require high-performance hardware resources.

5. Experimental Evaluation

This section presents the experimental evaluation results obtained from the maternal risk prediction models. Multiple supervised classification algorithms were evaluated and compared using several performance metrics in order to identify the most reliable and accurate approach for maternal healthcare risk classification.

The evaluated models include:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

The experimental analysis focuses on:

- quantitative performance comparison,

- model interpretability,
- classification reliability,
- generalization capability,
- and healthcare-oriented predictive insights.

5.1 Presentation of Results

All classification models achieved strong prediction performance on the maternal health dataset. The evaluation process included:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix Analysis
- Cross Validation Scores

The evaluation metrics were generated programmatically using Scikit-learn evaluation functions.

Logistic Regression Results

The Logistic Regression model demonstrated strong classification capability on the maternal health dataset.

<i>Metric</i>	<i>Value</i>
<i>Accuracy</i>	95.44%
<i>ROC-AUC</i>	0.9531
<i>Precision</i>	0.97
<i>Recall</i>	0.96
<i>F1-Score</i>	0.96

The confusion matrix showed:

- 90 correctly classified low-risk observations,
- 140 correctly classified high-risk observations,
- and only 11 total misclassifications.

The model achieved balanced precision and recall values, indicating stable binary classification performance.

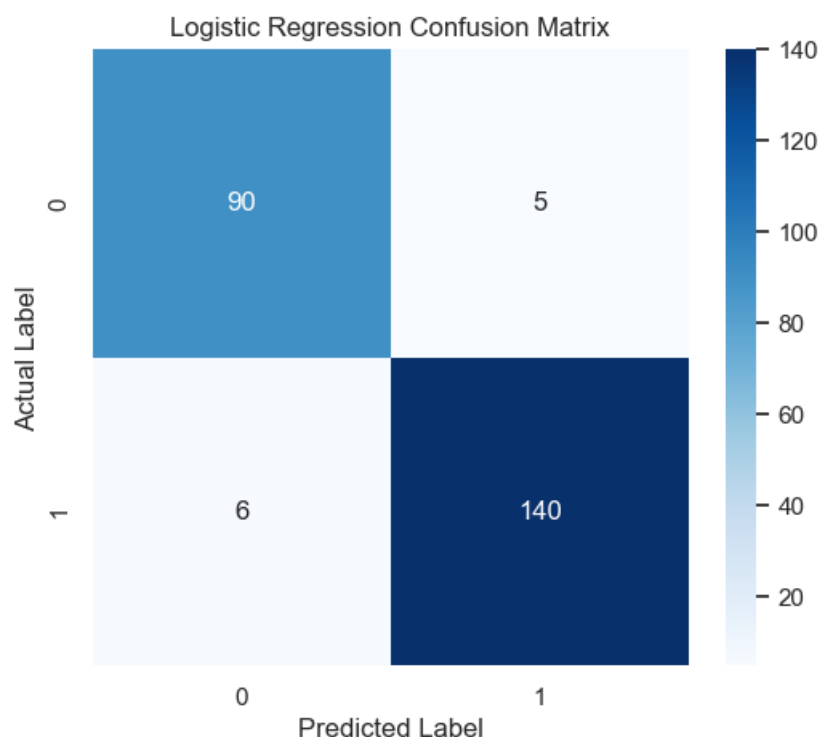


Figure: Logistic Regression Confusion Matrix

K-Nearest Neighbors (KNN) Results

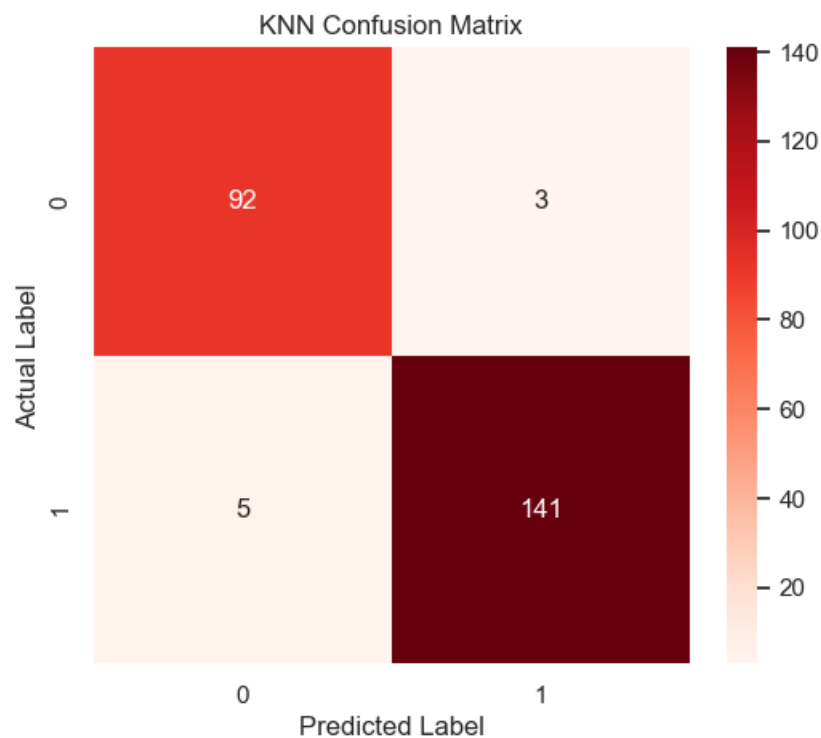
The KNN model produced highly competitive classification results after feature scaling.

<i>Metric</i>	<i>Value</i>
<i>Accuracy</i>	96.68%
<i>ROC-AUC</i>	0.9671
<i>Precision</i>	0.98
<i>Recall</i>	0.97
<i>F1-Score</i>	0.97

The confusion matrix demonstrated:

- 92 correctly classified low-risk observations,
- 141 correctly classified high-risk observations,
- and only 8 total classification errors.

The model successfully captured similarity-based relationships among maternal health profiles.



Support Vector Machine (SVM) Results

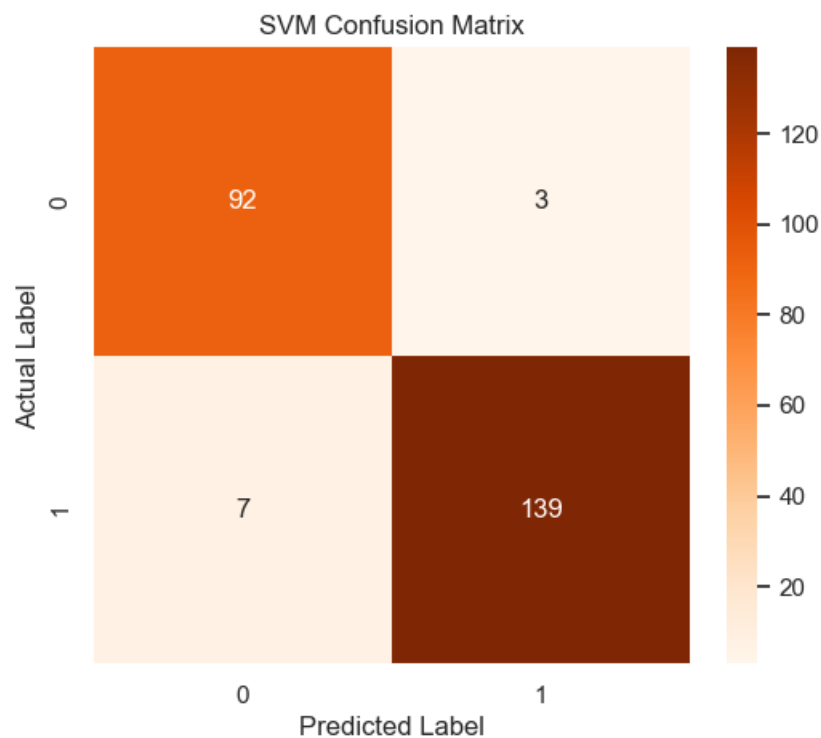
The SVM model achieved strong separation capability between maternal risk categories.

<i>Metric</i>	<i>Value</i>
<i>Accuracy</i>	95.85%
<i>ROC-AUC</i>	0.9602
<i>Precision</i>	0.98
<i>Recall</i>	0.95
<i>F1-Score</i>	0.97

The confusion matrix showed:

- 92 correctly classified low-risk observations,
- 139 correctly classified high-risk observations,
- and only 10 classification errors.

The model demonstrated strong margin-based classification performance and stable generalization capability.



Decision Tree Results

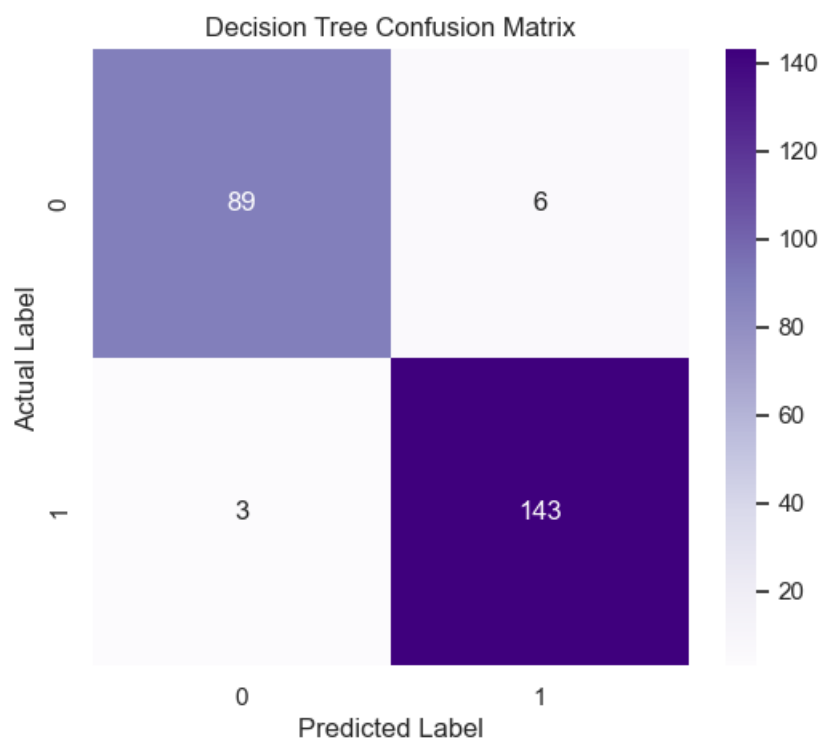
The Decision Tree model demonstrated effective nonlinear classification performance.

<i>Metric</i>	<i>Value</i>
<i>Accuracy</i>	96.27%
<i>ROC-AUC</i>	0.9581
<i>Precision</i>	0.96
<i>Recall</i>	0.98
<i>F1-Score</i>	0.97

The confusion matrix revealed:

- 89 correctly classified low-risk observations,
- 143 correctly classified high-risk observations,
- and only 9 total classification errors.

The model achieved particularly strong recall performance, indicating successful detection of high-risk maternal cases.



Decision Tree Confusion Matrix

Random Forest Results

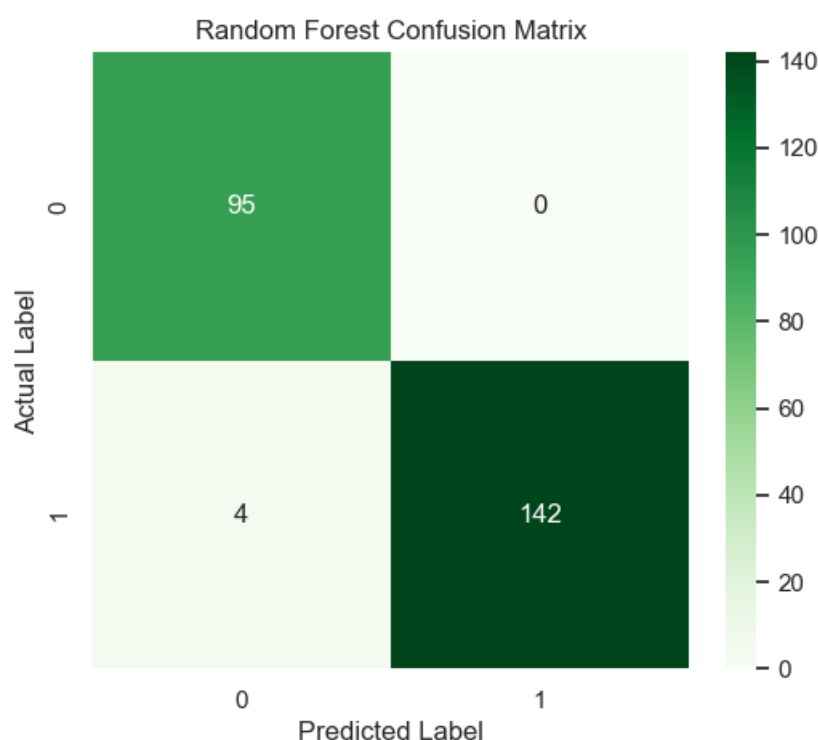
Random Forest achieved the strongest overall classification performance among all evaluated models.

<i>Metric</i>	<i>Value</i>
<i>Accuracy</i>	98.34%
<i>ROC-AUC</i>	0.9863
<i>Precision</i>	1.00
<i>Recall</i>	0.97
<i>F1-Score</i>	0.99

The confusion matrix showed:

- 95 correctly classified low-risk observations,
- 142 correctly classified high-risk observations,
- and only 4 total misclassifications.

The ensemble-based structure significantly improved prediction stability and reduced classification error.



Random Forest Confusion Matrix

5.2 Comparative Analysis of Models

The comparative evaluation demonstrated that all classification algorithms achieved highly successful maternal risk prediction performance. However, ensemble and tree-based approaches generally outperformed linear classification models.

The overall ranking of model performance is summarized below:

<i>Model</i>	<i>Accuracy</i>	<i>ROC-AUC</i>
<i>Random Forest</i>	98.34%	0.9863
<i>KNN</i>	96.68%	0.9671
<i>Decision Tree</i>	96.27%	0.9581
<i>SVM</i>	95.85%	0.9602
<i>Logistic Regression</i>	95.44%	0.9531

Random Forest achieved the highest overall accuracy and ROC-AUC performance, demonstrating superior classification capability on structured maternal healthcare data.

KNN and Decision Tree models also produced highly competitive results, indicating that both similarity-based and nonlinear decision-based learning approaches are effective for maternal risk prediction tasks.

5.3 Analysis of Discovered Patterns

The experimental findings revealed several medically meaningful relationships within the dataset.

The analysis demonstrated that:

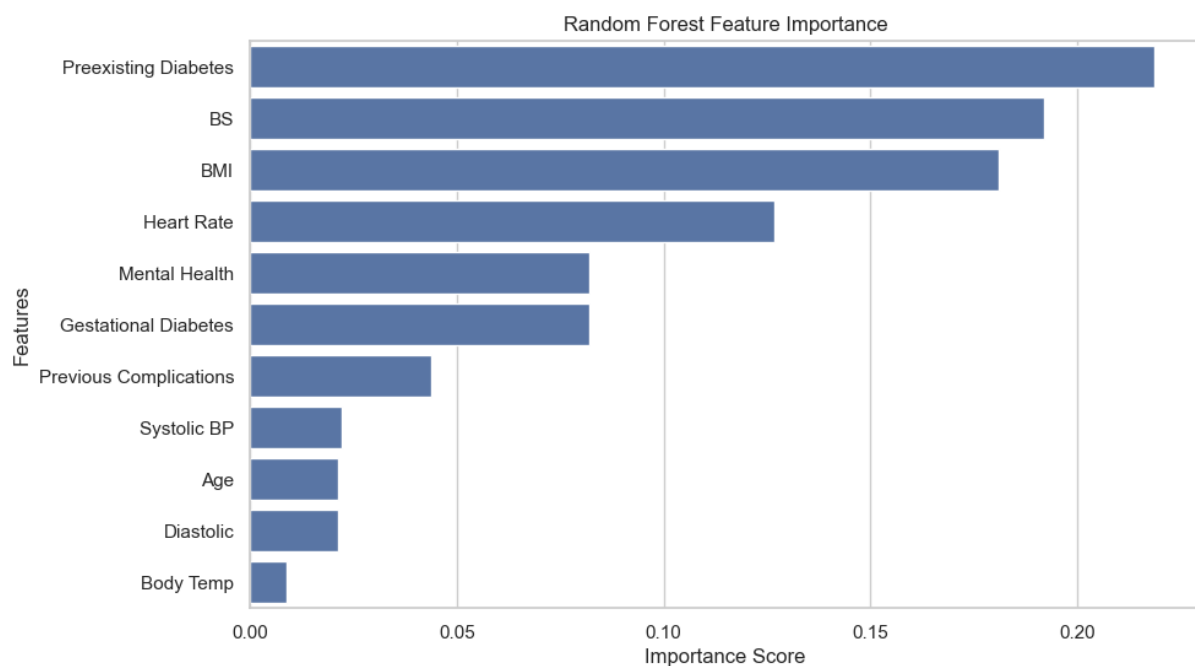
- high-risk maternal patients generally exhibit higher blood sugar levels,
- elevated systolic and diastolic blood pressure values are associated with increased maternal risk,
- BMI and diabetes-related variables strongly contribute to prediction performance,
- and previous pregnancy complications significantly influence maternal risk classification.

Correlation analysis and comparative visualizations further confirmed that blood sugar level, preexisting diabetes, BMI, and cardiovascular measurements are among the most influential predictors in the dataset.

The Random Forest feature importance analysis particularly highlighted:

- Preexisting Diabetes,
- Blood Sugar (BS),
- BMI,
- Heart Rate,
- and Previous Complications

as major contributors to maternal risk prediction.



Feature Importance

5.4 Performance, Interpretability, and Limitations

The evaluated classification algorithms demonstrated strong predictive capability and stable generalization performance. However, each model exhibits different strengths and limitations.

Logistic Regression

Advantages:

- interpretable probability outputs,
- computational efficiency,
- and simple implementation.

Limitations:

- limited capability in modeling complex nonlinear relationships.

KNN

Advantages:

- effective similarity-based learning,
- strong local pattern recognition.

Limitations:

- computationally expensive for larger datasets,
- sensitive to feature scaling and noise.

SVM

Advantages:

- strong generalization capability,
- effective margin-based classification.

Limitations:

- lower interpretability,
- higher computational complexity.

Decision Tree

Advantages:

- highly interpretable decision structure,
- successful nonlinear classification capability.

Limitations:

- higher overfitting risk when used individually.

Random Forest

Advantages:

- strongest overall classification performance,
- improved robustness,
- reduced overfitting risk,
- and feature importance analysis capability.

Limitations:

- lower interpretability compared to single decision trees,
- increased computational complexity.

Although the models achieved strong classification results, several limitations remain:

- the dataset size is relatively moderate,
- external clinical validation was not performed,
- and the dataset only contains binary risk categories.

Future studies may benefit from larger and more diverse healthcare datasets.

5.5 Error Analysis and Possible Improvements

Despite the strong overall performance, a small number of classification errors still occurred across all models.

The remaining errors may be caused by:

- overlapping medical characteristics between risk groups,
- noisy or incomplete clinical measurements,
- and nonlinear feature interactions that are difficult to separate perfectly.

In healthcare applications, false negative predictions are particularly important because undetected high-risk pregnancies may lead to severe clinical consequences.

Although Random Forest minimized total classification error, no model achieved completely perfect separation between all maternal risk categories.

Several potential improvements may further enhance prediction performance:

- applying advanced hyperparameter optimization techniques,
- using larger and more diverse maternal healthcare datasets,
- incorporating additional clinical indicators,

- applying ensemble boosting methods such as XGBoost,
- and evaluating deep learning-based healthcare classification approaches.

The experimental findings demonstrate that supervised data mining techniques can effectively support maternal healthcare analytics and provide reliable predictive performance for maternal risk assessment systems.

6. Conclusions and Future Work

This project focused on maternal risk prediction using supervised data mining classification techniques applied to a structured maternal healthcare dataset. The primary objective of the study was to evaluate the effectiveness of multiple classification algorithms in identifying low-risk and high-risk pregnancy conditions based on physiological and medical indicators.

Throughout the project, a complete data mining workflow was successfully implemented, including:

- exploratory data analysis (EDA),
- missing value handling,
- target encoding,
- feature scaling,
- model training,
- cross-validation analysis,
- and comparative performance evaluation.

Several supervised classification algorithms were evaluated, including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest.

The experimental findings demonstrated that all evaluated models achieved strong classification performance on the maternal health dataset. Among the tested algorithms, Random Forest achieved the strongest overall results with:

- 98.34% classification accuracy,
- 0.9863 ROC-AUC score,
- and the lowest overall classification error.

The results confirmed that ensemble-based classification techniques can effectively model complex healthcare relationships and provide highly reliable maternal risk prediction capability.

The analysis also revealed several medically significant predictors strongly associated with maternal risk levels, including:

- blood sugar level (BS),
- preexisting diabetes,
- body mass index (BMI),

- blood pressure measurements,
- and previous pregnancy complications.

These findings are consistent with previous healthcare studies and demonstrate the effectiveness of data mining approaches in extracting meaningful clinical knowledge from healthcare datasets.

One of the major strengths of this project is the comparative evaluation of multiple classification algorithms under the same preprocessing and evaluation framework. The use of multiple performance metrics, confusion matrix analysis, ROC-AUC evaluation, and cross-validation further improved the reliability and robustness of the experimental findings.

Another important strength is the interpretability of several evaluated models, particularly Decision Tree and Random Forest, which provide clinically meaningful insights regarding feature importance and decision-making behavior.

However, the study also contains several limitations. The dataset size is relatively moderate and may not fully represent large-scale real-world maternal healthcare populations. In addition, the dataset contains only binary maternal risk categories, whereas real clinical environments may involve more detailed and multi-level risk conditions. Furthermore, external clinical validation was not performed, which limits the direct medical applicability of the developed models.

During the project, several important lessons were learned regarding healthcare-oriented data mining workflows. The study demonstrated that preprocessing operations such as missing value handling and feature scaling significantly influence classification performance. The project also highlighted the importance of selecting appropriate evaluation metrics in healthcare applications, particularly recall and ROC-AUC, where false negative predictions may have serious clinical consequences.

Future studies may improve this work by:

- using larger and more diverse healthcare datasets,
- applying advanced hyperparameter optimization techniques,
- incorporating additional clinical variables and laboratory measurements,
- evaluating deep learning approaches,
- and implementing advanced ensemble boosting algorithms such as XGBoost or LightGBM.

Future research may also focus on real-time clinical decision-support integration and multi-class maternal risk prediction systems capable of providing more detailed healthcare recommendations.

In conclusion, the findings of this project demonstrate that supervised data mining classification techniques can successfully support maternal healthcare analytics and provide highly accurate maternal risk prediction performance. The developed models show strong potential for future healthcare-oriented predictive systems and clinical decision-support applications.

7. References

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